Question

M is an element. The high-valent oxide **A** of **M** is orange-yellow and can be obtained by the thermal decomposition of salt **B** containing **M** at 43.55% (retaining four significant figures). **A** serves as an important catalyst in industrial production. When **A** is fused with oxalic acid, it forms a deep blue compound **C** while releasing gas. **C** adopts a rutile structure at room temperature. The reduction of **C** with hydrogen yields oxide **D**, which contains **M** at 63.16% (retaining four significant figures). Further reduction of **C** produces the elemental form of **M**. The direct reaction of **M** with chlorine gas yields **E**, which is unstable in air. Under a CO atmosphere, **E** is reduced by Na in pyridine to form the highly symmetric anionic salt **F**. Upon acidification of **F**, a volatile blue-green crystal **G** is obtained. A similar low-temperature compound **H**, containing **M** at 23.25% (retaining four significant figures), belongs to the O_h point group.

From the following options, select the correct statement.

A. **B** has three more elements than **A**

B. In substance **D**, the ratio of the number of atoms of the element with the highest atomic count to that of the element with the lowest atomic count is greater than 2.

C. **F** molecular weight less than 200

D. **G** contains one more type of element than **H**.

E. The sum of the valence states of the metal elements in **C** and **E** is greater than 7.

F. In this problem, there are a total of 5 valence states for the metal elements among all substances.

Answer

Correct Answer: D

Detailed Explanation

This question provides hints in multiple places, and can be approached from the color of the substances or any mass fraction.

Based on the color of high-valent oxides, chromium trioxide and vanadium pentoxide can be initially screened. Among them, vanadium pentoxide is a more suitable industrial catalyst (sulfuric acid production) as required by the question. According to the mass fraction, if the metal is vanadium, the molecular weight of the salt is 116.97, corresponding to ammonium vanadate, while chromium does not have a corresponding one.

It can be inferred that A and B, A is V_2O_5;

CHECKPOINT 1 PTS

A is V_2O_5

B is NH_4VO_3; Thermal decomposition of ammonium vanadate is a common method for producing vanadium pentoxide, which verifies the correctness of the calculated result.

CHECKPOINT 1 PTS

B is NH_4VO_3

Based on the color, rutile structure, and common chemical knowledge, it can be inferred that C is VO_2;

CHECKPOINT 1 PTS

C is VO_2

Known as an oxide, based on the mass fraction and reduction conditions, it can be inferred that the ratio of vanadium to oxygen is 1:1.857, so D is V_7O_{13};

CHECKPOINT 1 PTS

D is V_7O_{13}

Based on common knowledge of elemental reactions, E is VCl_3;

CHECKPOINT 1 PTS

E is VCl_3

Reduced by carbon monoxide atmosphere and then by sodium, F is $NaV(CO)_6$;

CHECKPOINT 1 PTS

F is $NaV(CO)_6$

After acidification, based on common chemical knowledge and volatility, it can be known that G is $V(CO)_6$;

CHECKPOINT 1 PTS

G is $V(CO)_6$

Based on similarity, point group, and mass fraction, it can be known that H is $V(N_2)_6$; Here, the mass fractions of $V(CO)_6$ and $V(N_2)_6$ both match, but vanadium hexacarbonyl has already been used for the previous substance.

CHECKPOINT 1 PTS

H is $V(N_2)_6$

In summary, analyzing the options, it is easy to know that only D is correct. This question focuses on the correct deduction of all substances.

